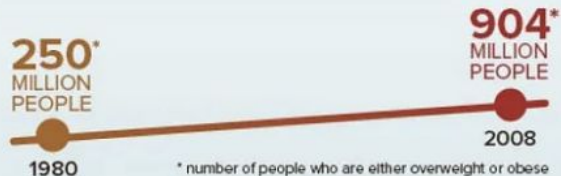




Predicting Obesity from Lifestyle Patterns

By: Anaya, Tsion, and Eden

THE WORLD IS GETTING FATTER



HOW DO I KNOW WHETHER I AM OVERWEIGHT?

Calculate your body mass index (BMI) using this formula

$$\text{BMI} = \frac{\text{weight (kg)}}{\text{height}^2 (\text{m}^2)}$$



OBESITY KILLS!

7 common diseases due to obesity:

- Arthritis
- Cancer
- Infertility
- Heart Diseases
- Back Pain
- Diabetes
- Stroke

OBESITY IS
KILLING
PREVENTABLE
THE WORLD

ABC TO OBESITY PREVENTION

SIMPLE RULES TO STAY IN SHAPE

A dopt New Healthy Habits



B alance Your Calorie Intake



C ontrol Your Weight Gain



Research Question

Which lifestyle factors are most strongly associated with obesity, and how effectively can these behaviors be used to predict whether an individual is obese or not, independent of their physical characteristics?

Dataset overview

2,111 individuals

Original target: 7-class
obesity label
(NO**Ob**eyesdad)

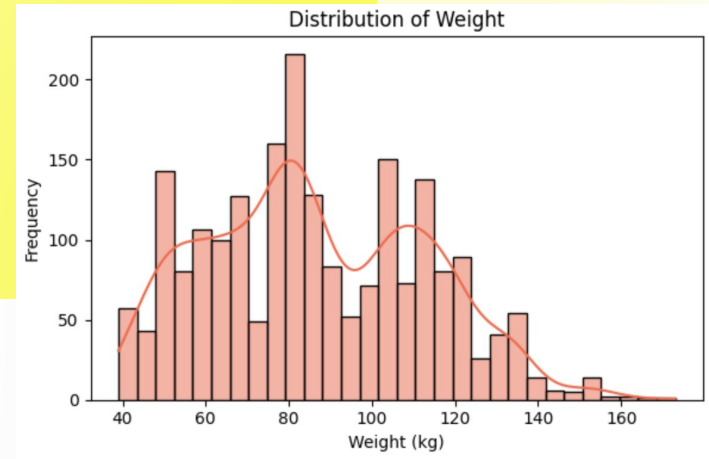
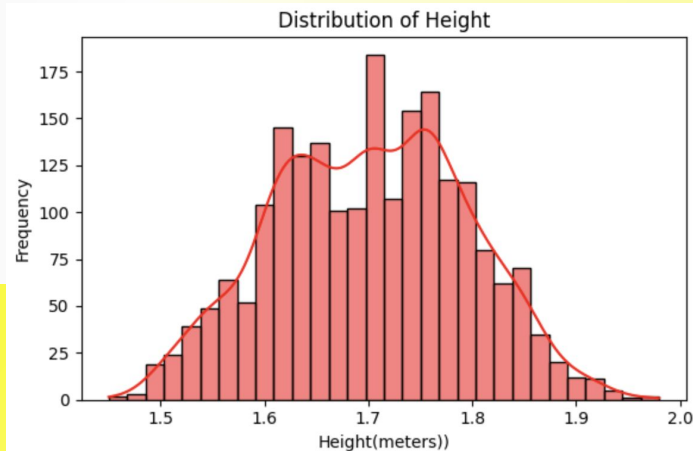
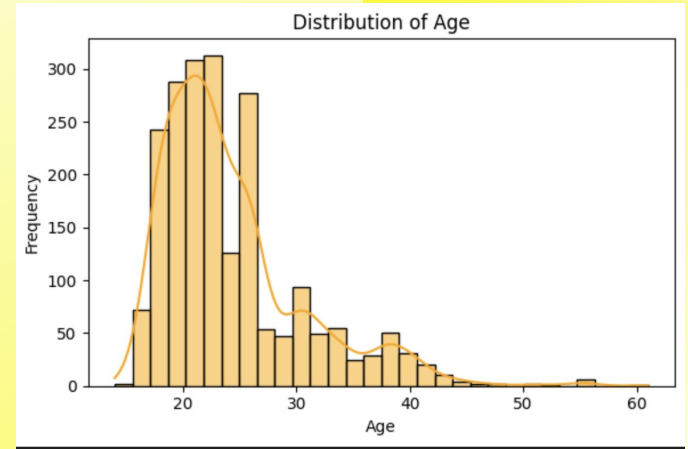
Lifestyle Features →

Source: Machine Learning
Class Repository

Column Name	Description
FAVC	Frequent consumption of high-calorie food (yes/no)
FCVC	Frequency of vegetable consumption (scale: 1 = low, 3 = high)
NCP	# of main meals consumed per day
CAEC	Frequency of eating between meals (e.g., Sometimes, Frequently)
SMOKE	Smoking habit (yes/no)
CH20	Daily water intake (scale: 1 = low, 3 = high)
SCC	Monitoring of daily caloric intake (yes/no)
FAF	Frequency of physical activity (in hours per week)
TUE	Daily time spent on technology (in hours)
CALC	Frequency of alcohol consumption (e.g., no, Sometimes, Frequently)
MTRANS	Primary mode of transportation (e.g., Walking, Public Transportation)

Exploratory Data Analysis

Among individuals classified as obese in this dataset (i.e., Obesity_Type_I, II, or III), the most common demographic profile is a 25-year-old male, approximately 1.72 meters tall, and weighing around 110 kg. These statistics reflect the average physical characteristics of people who are already obese.

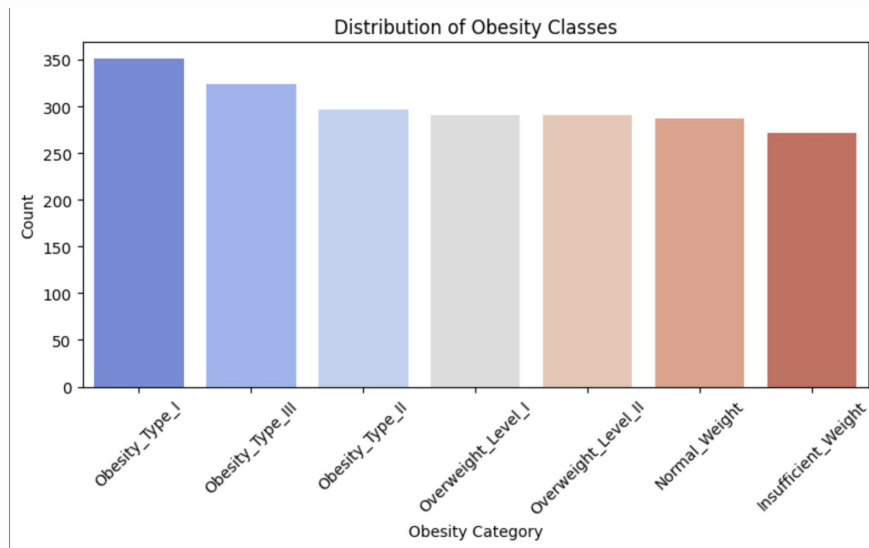


Why initial approaches failed

- Our initial approaches
 - Demographic Profiling of Obese Individuals
 - Not insightful, doesn't tell us why
 - Classifying between Types of Obesity
 - Model's performance was limited
 - Feature importance were misleading - many lifestyle variables had near-zero importance
 - Individuals in the obese group had similar shared habits → little variability

Ultimately, we did a Binary Classification , Obese vs. Not Obese, using the lifestyle behaviors. This provided a clearer and more useful prediction model.

Binary classification



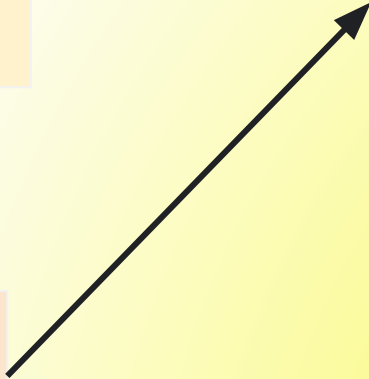
- Base rate and class distribution indicate that the three types of obesity account for 44% of the dataset and are evenly distributed
- Framed as a binary classification
 - Obese (Obesity_Type_I, II, III) vs Not Obese (Normal_Weight, Overweight, Insufficient_Weight)

Methodology

One-hot encoding for categorical features



Removed: Age, Gender, Height, Weight



Train/tune/test split: 80/10/10 (stratified)



Model: DecisionTreeClassifier. Hyperparameter tuned using GridSearchCV. 10-fold CV with 5 repeats



Evaluation metric: Balanced accuracy

Model Performance

Train Accuracy: 99.3%

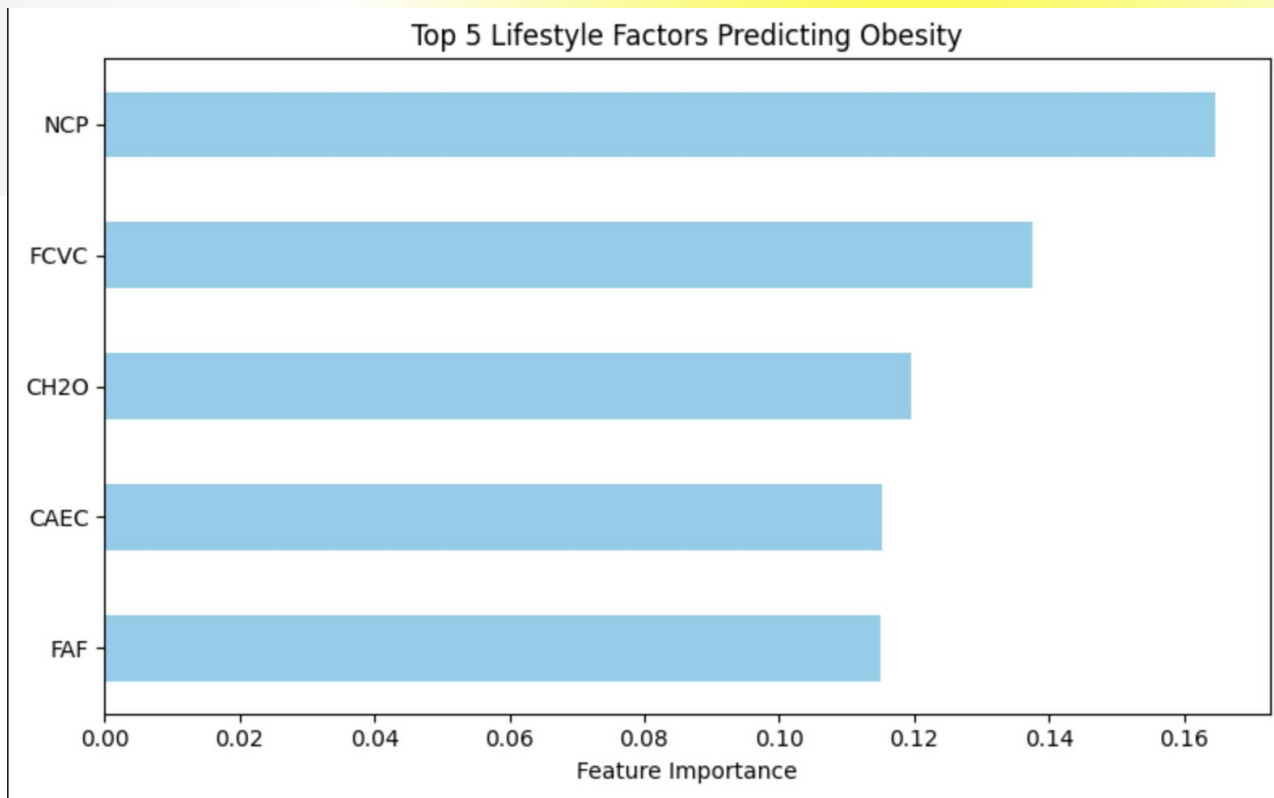
Tune Accuracy: 87.7%

Macro F1 Score: 0.88

Interpretation: Strong performance with no overfitting

Model predicts obesity without relying on physical traits

Feature Importance



NCP - # of main meals consumed per day

FCVC- Frequency of vegetable consumption (scale: 1 = low, 3 = high)

CH2O- Daily water intake (scale: 1 = low, 3 = high)

CAEC- Frequency of eating between meals (e.g., Sometimes, Frequently)

FAF- Frequency of physical activity (in hours per week)

Fairness and Interpretability

- No gender, age, weight, or height used in the final model
- Model focuses on modifiable behavior only
- Decisions made by the tree are transparent and explainable

Strengths & Limitations

Strengths:

- Focus on behavior → actionable insights
- High accuracy + interpretable model

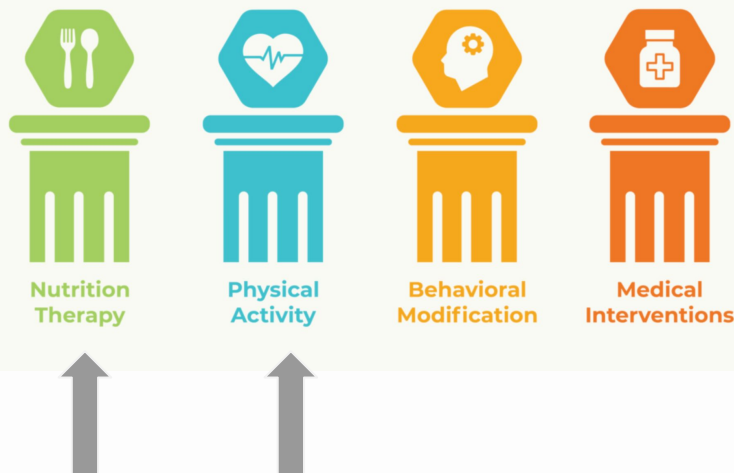
Limitations:

- Self-reported data may introduce bias
- Decision trees can be sensitive to noise

Future Work

- Use Random Forests for better stability and accuracy
- Validate on more diverse datasets
- Incorporate longitudinal data (e.g., food logs, fitness trackers)
- Extend to multiclass prediction when more data available

Final Takeaway



- Behavioral features can predict obesity with 88% accuracy
- Diet structure, hydration, and activity level are key signals
- This model shows potential for early, habit-based intervention strategies

Bibliography

<https://medium.com/@anitz44/big-people-big-problems-a9b683defcd1>

Dataset: course repository

<https://obesitymedicine.org/about/four-pillars/>